

Zain Ul Abidin

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Education	Sukkur IBA University Bachelor of Engineering (B.E.) in Computer Systems Engineering • Thesis: Machine Learning-Based Subsurface Defect Detection in CFRP Using Optical Pulsed Thermography • Supervisor: Dr. Junaid Ahmed [Thesis] [Code] • Cumulative CGPA: 3.80 / 4.00, Rank: 3/39	July 2026
Research Experience	Sukkur IBA University <i>Research Assistant / Independent Researcher</i> • Developed machine-learning and computer-vision pipelines for image-based detection and segmentation. • Applied image and signal processing techniques to real-world sensing problems. • Worked with deep learning models for visual recognition, segmentation, and reconstruction. • Explored mathematical and optimization-based approaches for intelligent sensing and perception. • Conducted experimentation, quantitative evaluation, and scientific writing.	2025–2026
Research Interests	• Machine Perception & 3D Understanding: 3D vision and depth estimation, pose and viewpoint estimation, geometric reasoning, scene understanding. • Learning & Physical Systems: Physics-informed machine learning, inverse problems in imaging, sensor fusion, signal-based perception. • Optimization & Autonomy: Mathematical optimization, active perception & sensor placement, uncertainty-aware decision making, autonomous robots and physical systems.	
Publications	Intelligent Perception & 3D Understanding [1] Habeeban Memon, Zain Ul Abidin, Junaid Ahmed. “Automated Depth Estimation and 3D Reconstruction of CFRP Subsurface Defects via Optical Pulsed Thermography.” <i>Transactions on Instrumentation and Measurement, IEEE</i> , 2026. [Preprint] [2] H. Memon, Zain Ul Abidin, J. Ahmed, A. Siddiqua. “Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning.” <i>5th International Conference on Computing, Mathematics and Engineering Technologies (iCoMET), IEEE</i> , pp. 1-6, 2026. [DOI] Multimodal & Sensor Fusion [3] Habeeban Memon, Zain Ul Abidin, Umair Ayaz Kamangar. “CMM-DSE: Cross-Modal Consistency-Guided Multimodal Fake News Detection via Dempster-Shafer Evidential Reasoning,” 2026. [Preprint] [4] Zain Ul Abidin, Habeeban Memon, Junaid Ahmed. “Three-Channel Thermographic Fusion for Architecture-Independent Defect Detection and Segmentation in Carbon Fibre Reinforced Polymer Composites Using Deep Learning.” <i>Engineering Applications of Artificial Intelligence, Elsevier</i> , 2026. [Preprint] Mathematical & Computational Perception [5] Zain Ul Abidin, Junaid Ahmed, Dong Wang, Guiyun Tian. “Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion Model.” <i>Nondestructive Testing and Evaluation, Taylor and Francis</i> , 2025. [Preprint] [6] Junaid Ahmed, Zain Ul Abidin, Guiyun Tian, Dong Wang. “Eddy Current Pulsed Thermography based Efficient Defect Detection in Metals via Keyframe Extraction and Accelerated Sparse Decomposition.” <i>NDT & E International, Elsevier</i> , 2025. [Preprint]	

Additional Publications

- [7] Zain Ul Abidin, Haseeb, H. Memon, M. Younis, J. Ahmed. "Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers." *International Journal of Multidisciplinary Conference Proceedings (IJMCP)*, Vol. 2(1), 2025. [\[DOI\]](#)

Research Projects	X-ray Prohibited Item Detection in Airport Inspection 2026 - Developed an automated computer vision system using deep learning architectures. - Segmented prohibited items within complex, highly cluttered X-ray security scans. [Code]
	KL-Divergence-Driven Online Sensor Relocation for Non-Stationary Barrier Coverage 2026 - Implemented sliding-window kernel density estimation and greedy sensor relocation. - Applied mathematical reasoning, optimization, and decision-making in dynamic environments to improve mean detection rate from 65.16% to 87.30%. [Code]
	AI-Based Android Controlled Vision Robot 2025 - Custom mobile robot utilizing ESP32-CAM for live video streaming. - TensorFlow Lite integration for real-time, on-device object detection. [Code]
Technical Skills	Mathematical Methods: Continuous & Combinatorial Optimization · Digital Signal Processing · Tensor Methods · Inverse Problems Machine Learning: Deep Learning · Physics-Informed ML · Supervised/Unsupervised Learning · Ensemble Methods Computer Vision: 3D Vision · Visual Perception · Image Processing · Semantic & Instance Segmentation Programming Languages: Python · C/C++ · MATLAB · Verilog Systems & Hardware: Embedded Systems (ESP32) · Edge Computing (TensorFlow Lite) · Hardware Synthesis (FPGA)
Conference Presentations	International Conference on Computing, Mathematics and Engineering Technologies (iCoMET 2026) Paper Presentation International Conference on Innovations in Science, Computing & Technology (ICISCT'25) Paper Presentation
Honors & Awards	Prime Minister's Laptop Award 2024 National merit-based award for academic excellence Sindh Educational Endowment Fund (SEEF) Scholarship 2022–2026 Fully merit-based undergraduate scholarship
	Mathematics for Machine Learning and Data Science Specialization Coursera Machine Learning Specialization Stanford / Coursera Deep Learning Specialization: Advanced AI, Hands-on Lab Udemy
References	Dr. Junaid Ahmed Bhatti, Assistant Professor, Sukkur IBA University Dr.-Ing. Kashif Hussain, Assistant Professor, Sukkur IBA University Dr. Ghulam Abbas Lashari, Assistant Professor, Sukkur IBA University